OnTrek

A smartwatch application to simplify your hikes

Human Computer Interaction on the Web
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Sapienza University of Rome

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Problem statement

Inadequate Trail Markings

Sometimes mountain trails lack sufficient or clear signage, making the hike difficult and increasing the risk of getting lost or straying into dangerous terrain.

Losing the Trail Increases Risk

When hikers lose the path, they waste precious time trying to reorient themselves. In mountain environments, where weather conditions can change rapidly, this delay can significantly increase the danger.

Need finding: Questionnaire results

Why hikers leave the trail

- **60.6%**: Poorly visible trail signs
- **15.2%**: To admire the view / take photos
- **12.1%**: To take a break
- Other minor reasons: distraction, wrong advice, trail opening into the forest

How they found it again

- **75.8%**: By retracing their steps
- **30.3%**: Using a smartphone app
- **18.2%**: Asking other hikers
- **15.2%**: Using a paper map
- **3%**: Someone came to pick them up

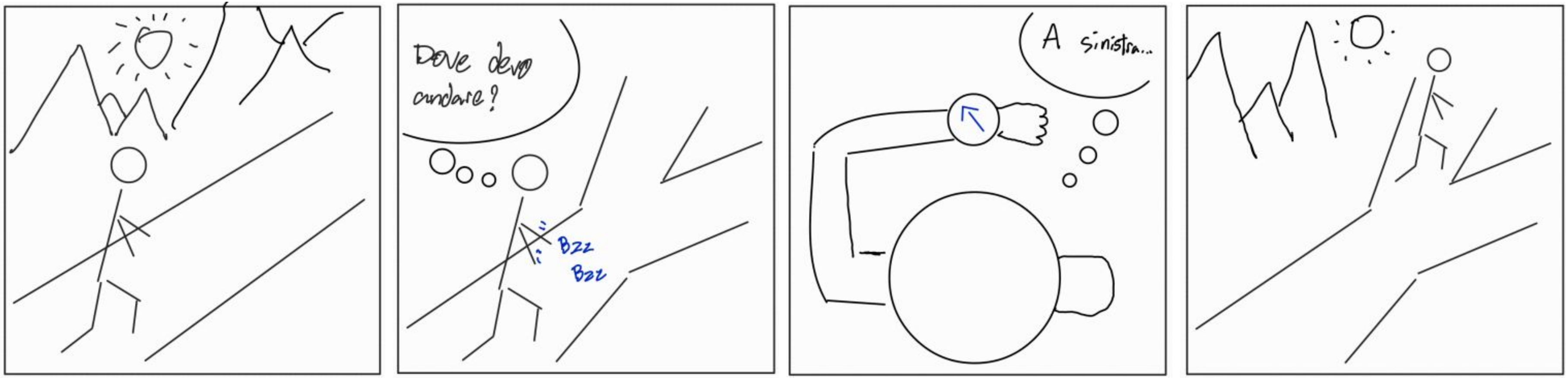
Time needed to find the trail

- **48.5%**: 10–30 minutes
- **33.3%**: 5–10 minutes
- **6.1%**: 0–5 minutes
- **6.1%**: 30–60 minutes
- **6.1%**: Over 60 minutes

Identified Needs

- Clear directions
- Stay on track
- Help requests

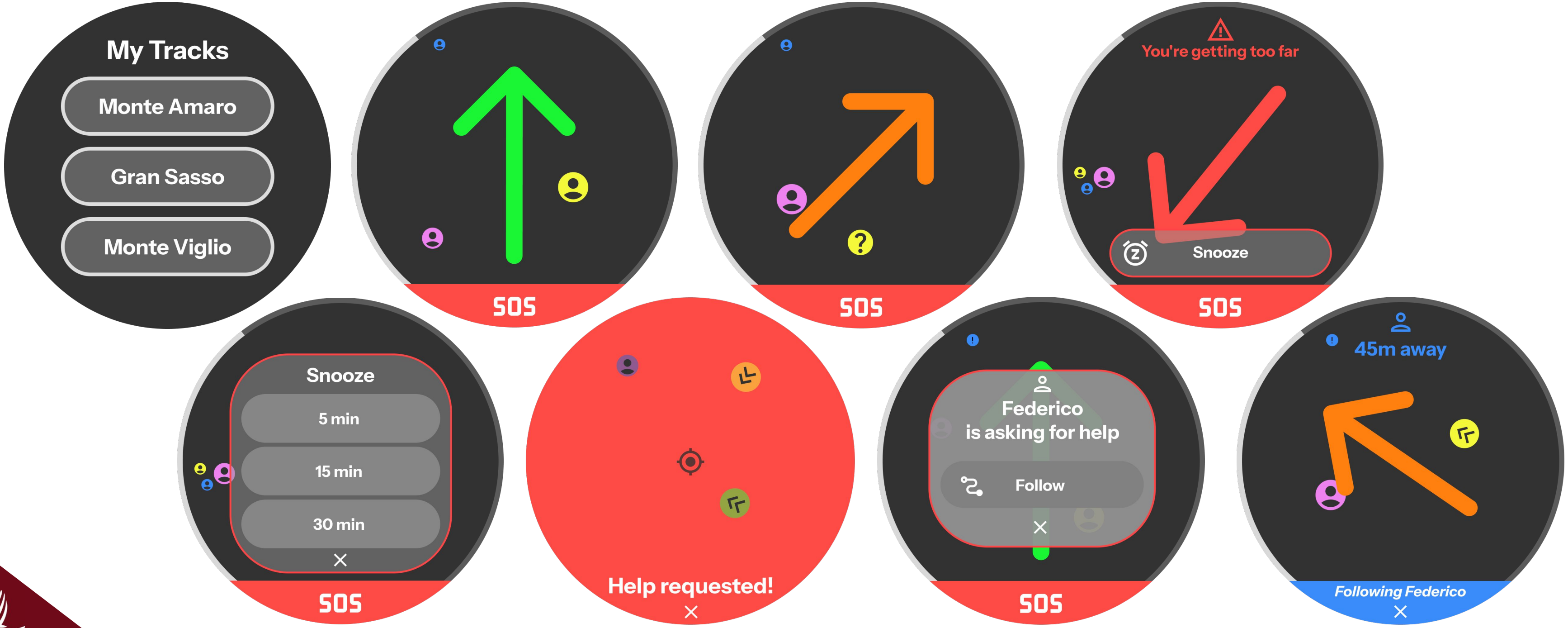
Storyboard - Follow the planned path



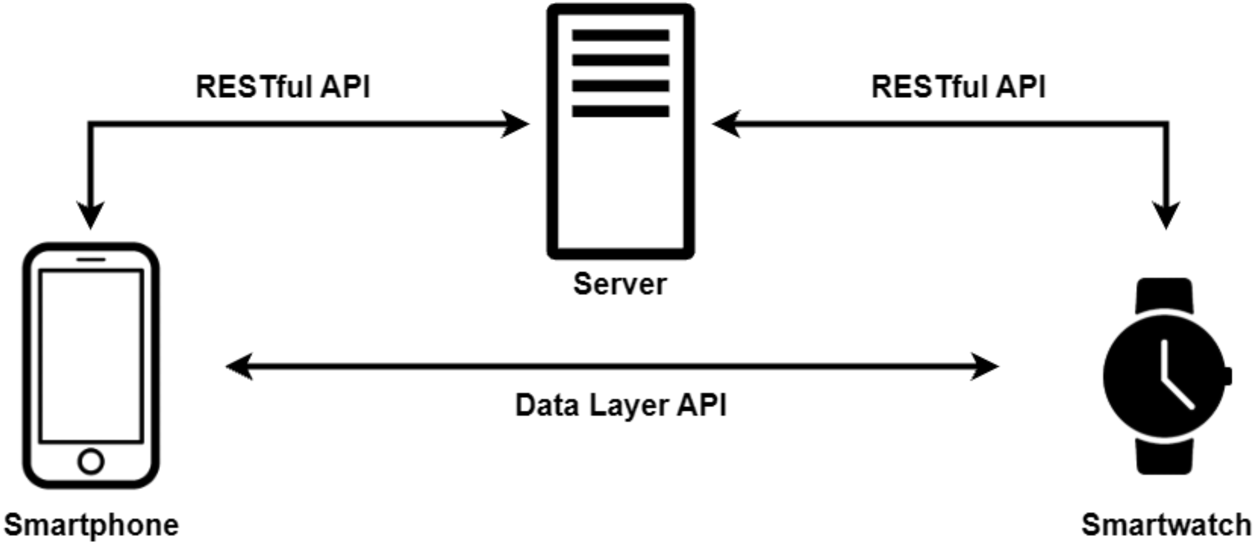
Storyboard - Off track alert

Storyboard - Directions to user who needs help

UI prototypes



Architecture overview



Backend

Used Technologies:

- Golang - Gin Framework
- Mapbox
- GORM
- SQLite

Core functionalities:

- User profile, hike groups and gpx files management
- Real time processing of group members’ location
- SOS assistance request management

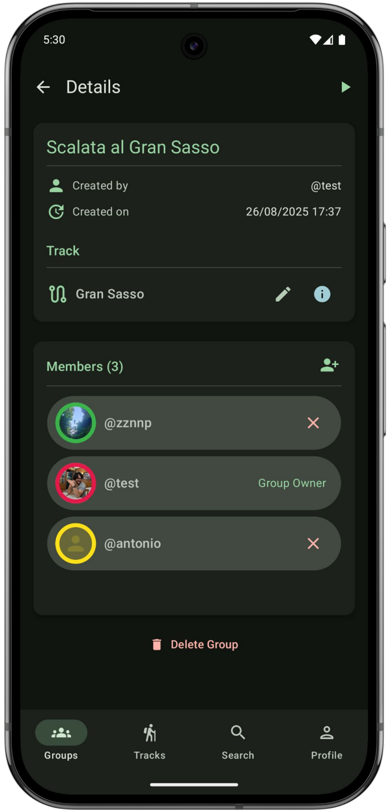
Frontend - Mobile

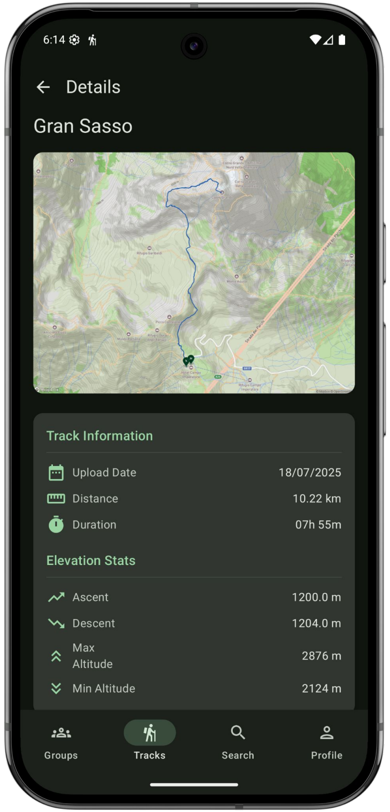
Used Technologies:

- Kotlin
- Android SDK
- Jetpack Compose

Core functionalities:

- Track upload (GPX files)
- Groups & friends management
- Profile management
- Authentication





Frontend - Wearable

Used Technologies:

- Kotlin
- Android SDK
- Jetpack Compose

Core functionalities:

- Directions while hiking
- Alerts when off track
- Request help from friends
- Follow needy friends

Hike modalities:

- With groups
- Alone

Direction based UI

Unlike traditional map-based applications, OnTrek interface avoids clutter and focuses on immediate, directional guidance, enhancing usability in outdoor environments.

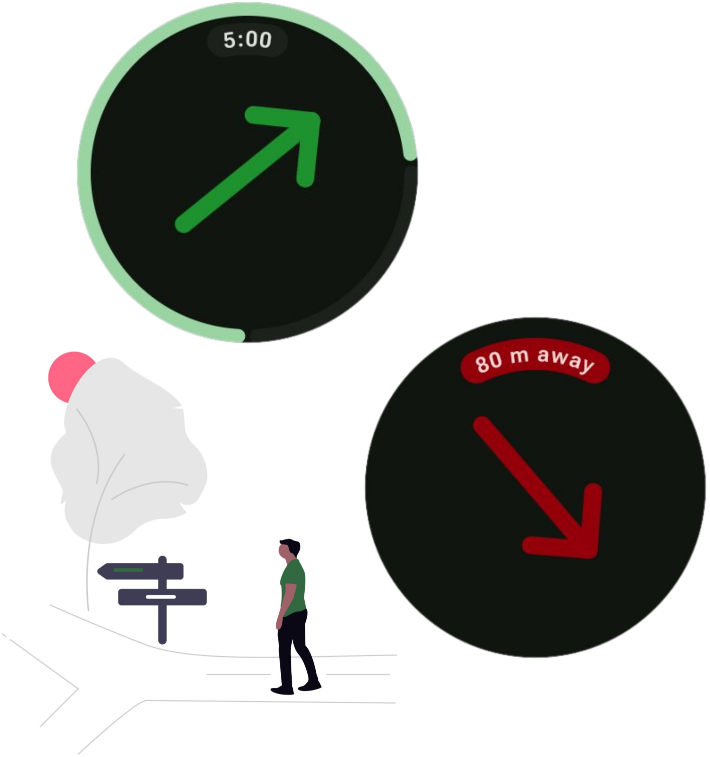
Directions while hiking alone

The user will see:

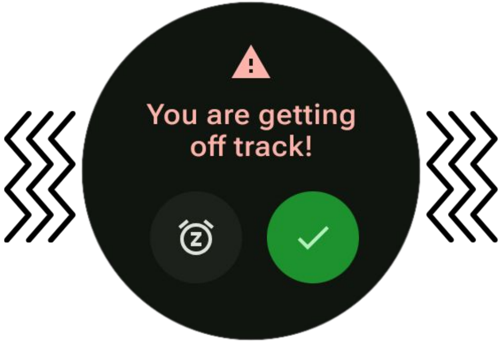
- An arrow always pointing the right direction that changes color based on the distance from the track (from green to red)
- A progress bar that represents the distance travelled compared to the total distance

How it is done:

- The trail is represented as a ordered list of geographical points, called waypoint
- The smartwatch GPS and magnetometer are used to get the location and orientation of the user
- The user location is used to compute the next point of the trail and the distance from the trail



Off track alert



When the user get far from the track the smartwatch will **vibrate** and show an **alert**

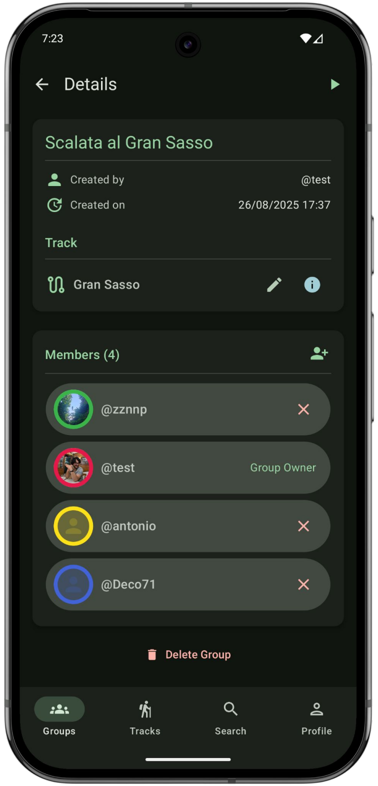
The user can choose to **close** the dialog or **snooze** the notification for a predefined period of time if they deliberately left the trail

Closing the alert the **distance** from the trail will be shown and the arrow will indicate the direction of the **nearest point** of it

Group hike

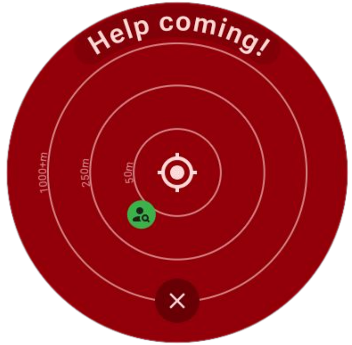
The hike can also be done in a group:

- All the group members are shown as a circle in a radar-like interface to represent their location
- Each member has an own color that is reflected on the mobile interface to distinguish them
- The user can also send help request to other members from the SOS button at the bottom



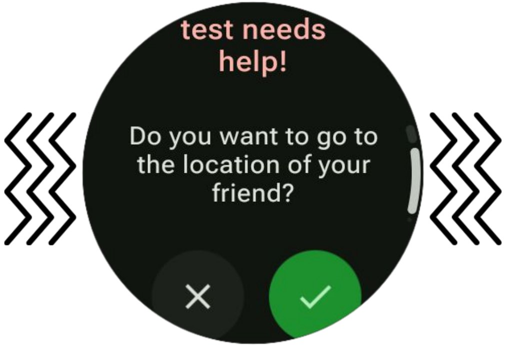
Send help request

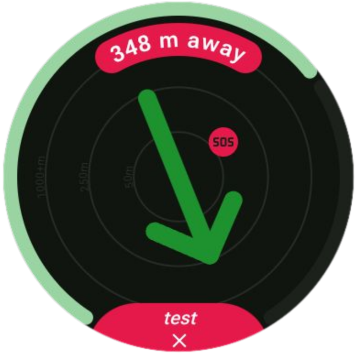
- The user must press and hold the SOS button for 5 seconds while the watch vibrate strongly and the screen progressively turn red
- The aim of the vibration is to give the user an haptic feedback in case they pressed the button by mistake
- If the user successfully perform this operation the app will navigate to the SOS screen
- Here they will see only the friends who accepted the request on the radar



Receive request and follow friend

- At the other end, all the other members of the group will receive an alert that informs them of the help request
- They can accept or refuse to go to the location of the user that request help for providing support
- If they accept the request:
    - the interface will change color to match the user's one
    - the arrow will indicate the path to their location following the hiking trail
    - the progress bar will indicate the distance travelled toward the user
    - the remaining distance in meters is shown at the top
- If the needy user is out of the track the arrow will direct to the nearest point of it and then indicate the direction of the user





Main challenges

- **Waypoint navigation**
    - Problem: determining which waypoint to direct users toward while following a track
    - Solution: adaptive distance-based algorithm that calculates the most appropriate target waypoint based on user location and previously targeted waypoints
- **Energy consumptions**
    - Problem: the smartwatch app drain the battery
    - Solution: stop the magnetometer when the screen is off
- **Mobile-watch interactions**
    - Problems: authentication and start tracks from mobile
    - Solution: Android SDK data client API

Evaluation during development

Lab tests:

- With Android Studio emulators
- Performed by us

Field tests:

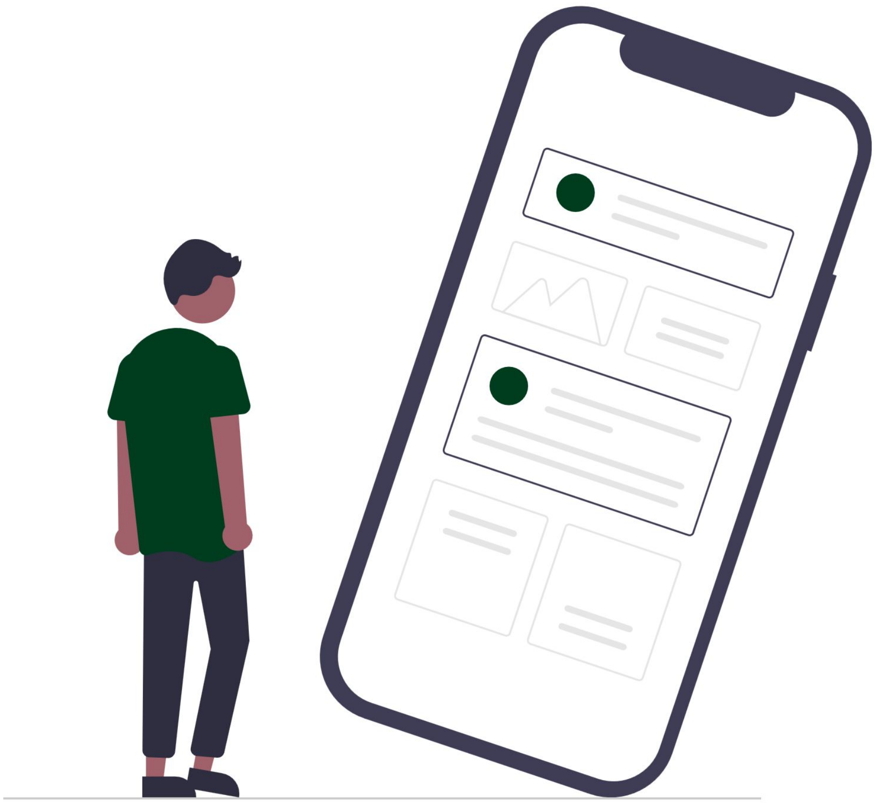
- On street/mountain trails
- Performed by us and ten external users

On field tests structure

Think aloud

We asked the users to perform these tasks on the mobile app:

- Add a new track
- Send a friend request
- Create an hiking group, add the relative track and friends
- Connect the smartwatch and start the hike group



On field tests structure

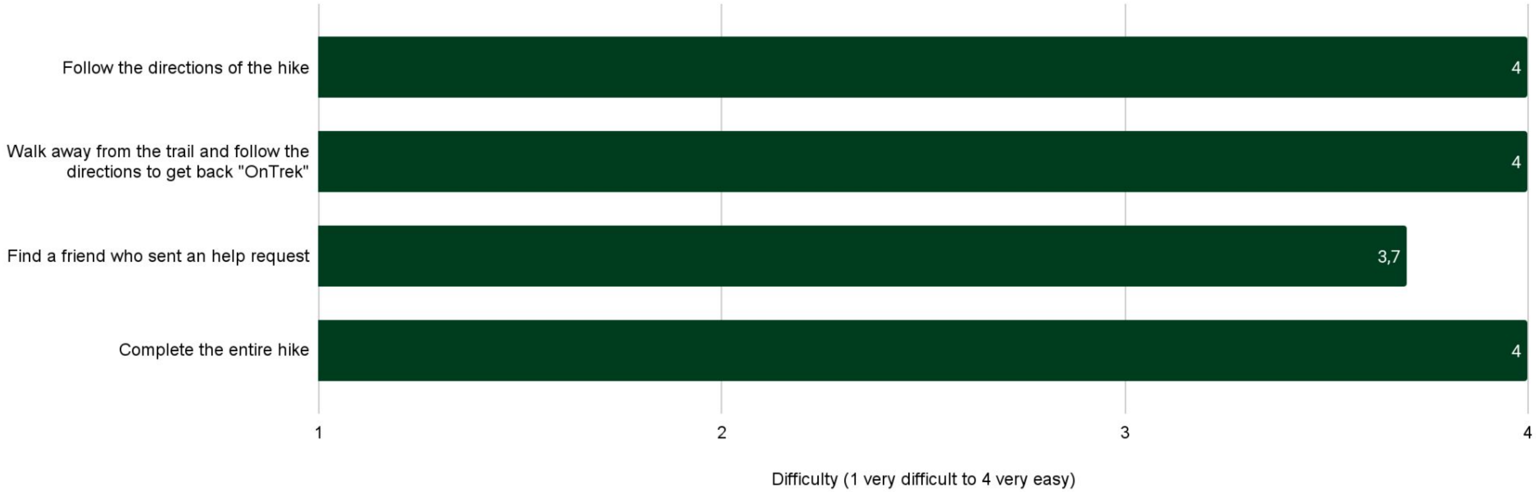
Think aloud

We asked the users to perform these tasks on the smartwatch app:

- Follow the directions of the hike
- Walk away from the trail and follow the directions to get back ”OnTrek”
- Walk away from the trail and snooze the notification for the desired time
- Send a help request
- Find a friend who sent an help request
- Complete the entire hike

On field tests - Results

As test observers, we evaluated each task performed by each user in a scale from 1 (very hard) to 4 (very easy). These are the collected average results of the most critical tasks performed by 10 users:



User feedbacks

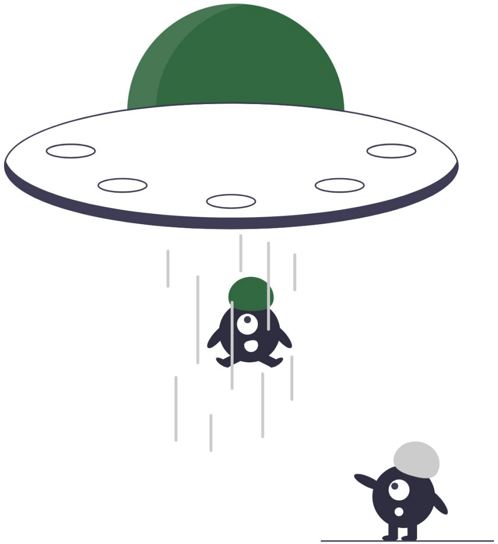
What did the users like the most:

- Direction accuracy
- Off-track notification
- Simple and uncluttered UI
- Distance progress bar

What did the users want us to improve:

- Arrow animation
- Homepage feed (similar to social media) with in-app notifications for the mobile app
- GPX editor in the mobile app
- Hike history with statistics

Future Improvements



- **Record hike statistics:** measure statistics about ongoing hikes
- **Track editor:** include a GPX editor in the mobile application to simplify the creation of new tracks
- **Fall detection (*work in progress*):** send an automatic help request when a user falls
- **Enhance connectivity:** integrating opportunistic networking techniques or satellite-based communication for areas with limited or no cellular coverage.

Fall Detection

Used sensors:

- Accelerometer(m/s²)
- Gyroscope(rad/s)

Training and preprocessing:

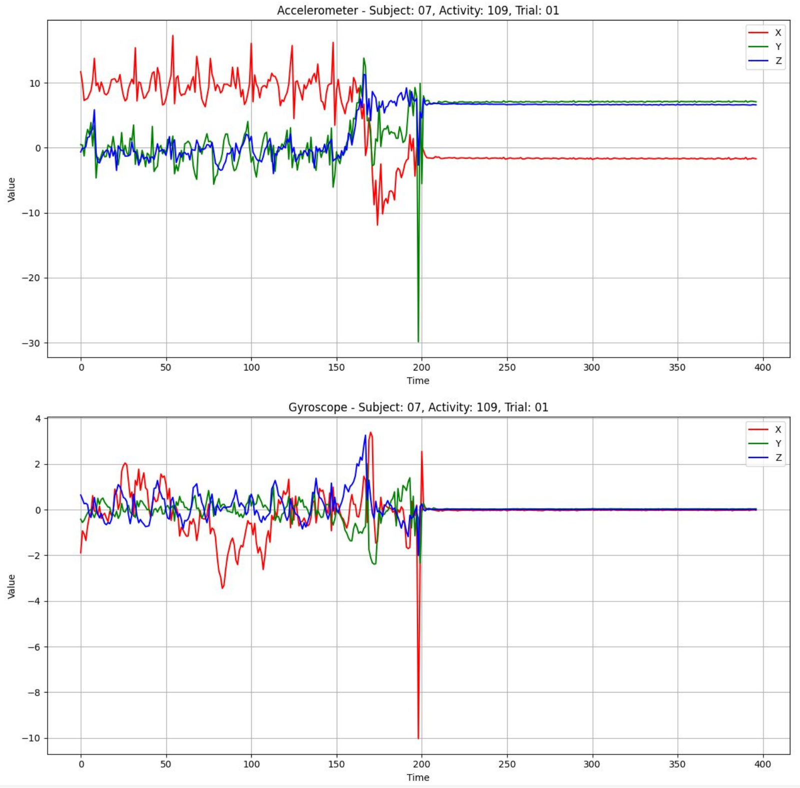
- Dataset: FallAllD
- Denoising: Kalman filter

Model:

- Coarse-fine CNN and GRU Networks
- A dedicated LSTM branch for processing gyroscope data

Deploy:

- The model was trained with data acquired at waist level, so smartphone is used for data acquisition and forwarding.
- If a fall is detected, a message is sent to the smartwatch.



Fall Detection

- When smartwatch receives the fall detected message from the mobile an alert is opened
- The user has **30 seconds** to tell if they are ok or not
- A progress bar will decrease along with the time left
- If the user responds that they are ok, the app will navigate back to the direction screen
- Otherwise, if the user is not ok or the time reaches the zero a **help request will be sent** automatically
- This mechanism introduces an implicit interaction modality: **inaction itself is interpreted as a potential emergency condition**, triggering the alert procedure

Thank you for the attention!